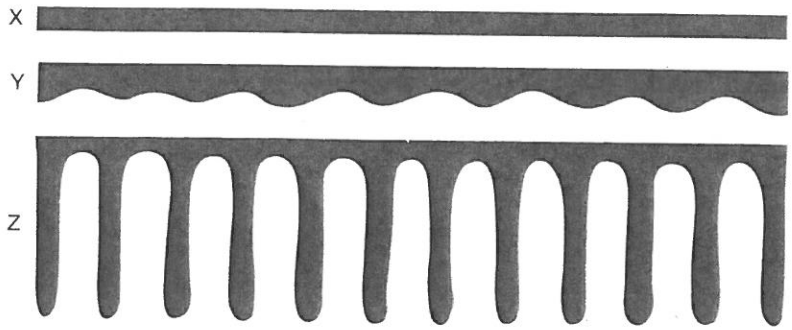
3.4 Investigating villi

Science inquiry skills

FOUNDATION | STANDARD | ADVANCED

Evaluating | Processing & Analysing

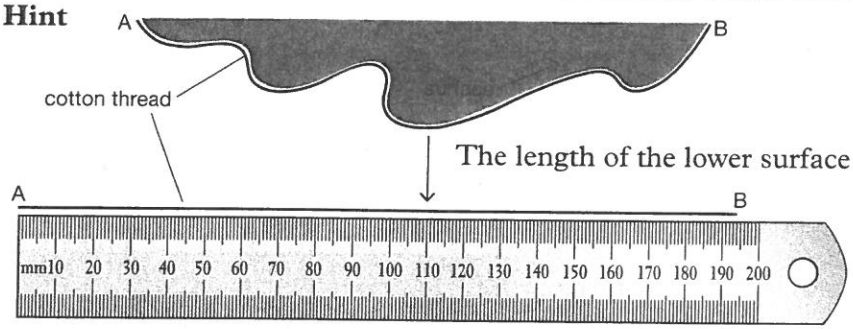
The following diagrams represent three different surfaces X, Y and Z.



1 Use cotton thread or fine string and a ruler to measure the length of the lower side of each surface. The Hint box below shows you how to do this.

Length of surface X _____ Y _____ Z _____

Hint



2 If each centimetre of surface absorbs 5 mL of digested material every 10 minutes, how much digested material is absorbed in one hour by each of the surfaces X, Y and Z. A worked example is provided to help you with your calculations.

X _____

Y _____

Z _____

Worked example:

- If the surface length is 10 cm
- And 5 mL is digested per cm
- Then 50 mL (10 cm × 5 mL) is digested every 10 minutes

As there are 60 (10 min × 6) minutes in an hour

- Then over an hour the total amount digested is 50 mL × 6 = 300 mL

3 Compare the effectiveness of surfaces X, Y and Z in digesting food.

_____

_____

4 Explain why it is an advantage to have villi lining the small intestine.

_____

_____

RATE MY UNDERSTANDING
Shade the face that shows your rating